



Institutional Pressure and Organizational Support: Configurational Pathways to Mobile Government Service Capacity in Chinese Provinces

Guangdong Wu¹, Qingwen Li¹, Zhibin Hu^{2*}, Xianbo Zhao³, Junlong Huang⁴

¹ School of Public Policy and Administration, Chongqing University, 400044 Chongqing, China

² School of Public Administration and Policy, Shandong University of Finance and Economics, 250002 Jinan, China

³ School of Engineering and Technology, Central Queensland University, NSW 2000 Sydney, Australia

⁴ Department of Ecology and Evolutionary Biology, University of Toronto, M5S 3B2 Toronto, Canada

* Correspondence: Zhibin Hu (hzbmail1023@163.com)

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Abstract: Digital public service delivery is an increasingly important component of local governance, yet substantial disparities persist in the capacity of provincial governments to provide effective mobile government services. Understanding how institutional environments interact with internal organizational resources is therefore essential for explaining uneven digital service performance across jurisdictions. This study investigates the configurational mechanisms through which institutional pressure and organizational support shape mobile government service capacity across Chinese provinces. An analytical framework combining coercive, mimetic, and normative pressures with technical, financial, and leadership support was developed, and fuzzy-set qualitative comparative analysis (fsQCA) was conducted using data from 31 provincial governments. The results showed that no single antecedent condition was necessary for high mobile government service capacity. Five configurations leading to high capacity were identified and grouped into three pathways: a coercive-pressure-led pathway, a mimetic/normative-pressure-driven pathway, and a three-pressure-linkage pathway. The overall solution consistency and coverage reached 0.885 and 0.631, respectively. Technical support was present across all five high-capacity configurations, while the joint absence of technical support and mimetic pressure characterized the core conditions associated with low capacity. Cross-pathway comparisons further revealed potential substitution between leadership support and combinations of normative pressure and financial support under different institutional conditions. The findings demonstrate that strong mobile government service capacity emerges from multiple combinations of external institutional pressures and internal organizational support rather than from any single governance condition. The study extends configurational explanations of digital governance and provides practical insights for provincial and local authorities seeking to align institutional demands, organizational resources, and digital public service provision.

Keywords: Digital governance; Mobile government services; Digital public service delivery; Institutional pressure; Organizational support; Fuzzy-set qualitative comparative analysis

1 Introduction

The expansion of digital government services has become an important policy-driven process through which central and local authorities seek to improve public service delivery and administrative coordination [1, 2]. As information technologies become increasingly networked and intelligent and public expectations evolve within more complex governance environments, public administration is shifting toward platform-based and digitally mediated forms of service provision [3]. Within this transformation, provincial governments have increasingly relied on digital platforms to reorganize service processes, improve administrative responsiveness, and develop more accessible channels for interaction between government agencies and service users [4]. The transition from offline to online workflows and from fragmented departmental provision to more coordinated administration has therefore made mobile government platforms an increasingly important component of integrated public service delivery and local governance capacity [5]. Despite substantial progress in digital platform development, considerable differences remain among jurisdictions in their ability to coordinate online and offline services, support cross-regional transactions, and meet

increasingly standardized requirements for government service provision [6]. These persistent differences suggest that technological development alone cannot adequately explain why some provincial governments achieve stronger mobile public service capacity than others.

Explaining this uneven performance requires attention to both the institutional environments in which local governments operate and the organizational resources through which policy requirements are translated into service capacity. During policy implementation, local governments are simultaneously embedded in the institutional logics of the local state, the administrative system, and bureaucratic organizations, and these overlapping logics can generate different priorities and courses of action [7, 8]. In a pressure-based governance system, national digital-service objectives are transmitted through hierarchical administrative channels, but their implementation ultimately depends on how local governments interpret these requirements and mobilize the resources available within their jurisdictions. As the principal actors responsible for translating higher-level policy objectives into operational arrangements, local governments formulate implementation strategies, coordinate administrative actors, and allocate resources to support digital public service provision [9]. Such responses may involve investments in digital infrastructure, dedicated financial resources, organizational coordination, and sustained leadership attention. At the same time, policy mandates are not the only external forces shaping these decisions. Peer competition, professional norms, and changing public expectations can create additional pressures, producing an institutional environment in which multiple demands interact rather than operate independently [10]. The capacity to deliver mobile government services should therefore be understood not simply as a technological outcome, but as the product of interactions between external institutional conditions and internal organizational responses.

Existing digital government research provides important explanations of institutional and organizational change, but the interaction between these two dimensions remains insufficiently understood. A substantial body of scholarship has examined digital governance within hierarchical institutional arrangements [11], investigated differences in local government digital transformation from the perspective of political cognition and organizational behavior [12], and developed broader models of e-government transformation and service delivery [13]. These studies establish the importance of institutional context, political incentives, and organizational characteristics, yet much of the literature continues to examine individual determinants or broad institutional environments separately. As a result, less is known about how different forms of institutional pressure combine with specific types of organizational support to produce divergent service capacities across jurisdictions. This limitation is particularly important in provincial digital governance, where similar national policy objectives may generate markedly different outcomes because local governments face different competitive environments, public expectations, fiscal conditions, technical resources, and levels of leadership attention. Institutional pressures provide the external context within which policy implementation occurs, while organizational cognition and response capacity determine how these pressures are interpreted and translated into concrete action [14]. Examining either side independently therefore risks overlooking causal interdependence, equifinality, and the possibility that different combinations of institutional and organizational conditions may lead to similarly high levels of mobile government service capacity.

Against this background, this study investigates the configurational pathways through which institutional pressure and organizational support shape mobile government service capacity across Chinese provinces. Drawing on neo-institutional theory, an integrated analytical framework is developed that connects three forms of external institutional pressure—coercive, mimetic, and normative pressure—with three dimensions of internal organizational support—technical, financial, and leadership support. Rather than estimating the isolated net effect of each condition, the study adopts a configurational perspective to examine how these conditions operate jointly across different provincial governance contexts. Specifically, the analysis addresses three related questions: (1) how do different forms of institutional pressure combine with organizational support to generate high mobile government service capacity; (2) what alternative configurations can produce comparable high-capacity outcomes; and (3) how do institutional pressures and organizational support substitute for or reinforce one another under different local conditions? By addressing these questions, the study extends research on digital governance from individual determinants toward the joint institutional and organizational mechanisms underlying differentiated public service performance. It also provides a place-sensitive explanation of why provincial governments operating within the same national institutional framework can follow different routes to effective mobile service provision, with practical implications for aligning institutional demands, organizational resources, and digital public service delivery across heterogeneous local governance contexts.

2 Literature Review and Theoretical Framework

2.1 Determinants of Mobile Government Service Capacity

Mobile government services are digitally mediated public services delivered through mobile-enabled digital platforms, representing an extension of e-government systems and a transformation of public service delivery processes [4]. One strand of research emphasizes organizational determinants, particularly the institutional structures, administrative arrangements, and leadership conditions that shape governments' capacity to adopt and sustain digital services. Ioannou et al. [15], for example, examined the relationship between organizational institutions and

e-government development, while Al-Khayari et al. [16] demonstrated the importance of leadership in supporting e-government adoption and performance. A second strand adopts a technology-driven perspective, treating technological capability as a fundamental condition for the effectiveness of e-government and smart government systems [17]. A third focuses on users and public demand, regarding citizens' expectations, participation, and service needs as important forces shaping digital government development. This perspective is reflected in the examination by Venkatesh et al. [18] of demand-related mechanisms in digital government construction and in the citizen-centered assessment by Singh et al. [19] of e-government portal effectiveness. Beyond these single-factor perspectives, an increasingly important body of research adopts more integrative approaches. Some studies employ established frameworks such as the Technology-Organization-Environment model to explain e-government adoption through technological, organizational, and environmental conditions [20, 21], whereas others draw on socio-technical theory to construct multidimensional explanations of online government service effectiveness [22]. Related studies have further conceptualized political and technical rationality as broader dimensions of online service capacity [23] or developed demand-pressure-support frameworks to explain variations in local governments' capacity to provide internet-based public services [6].

This literature provides a substantial theoretical and empirical foundation for understanding digital public service capacity, but two limitations remain particularly relevant. First, many existing studies identify technological, organizational, institutional, or demand-side determinants but continue to treat them primarily as separate explanatory dimensions or as additive components within predefined analytical frameworks. Such approaches are valuable for identifying relevant factors, yet they provide a more limited account of how conditions may reinforce, compensate for, or substitute for one another across different governance contexts. Theoretical development therefore requires greater attention to the interdependence of institutional environments and organizational responses rather than simply expanding the number of explanatory variables. Second, empirical research has established numerous relationships between individual determinants and e-government outcomes, but less attention has been devoted to explaining why governments operating under broadly similar national policy arrangements can nevertheless develop substantially different levels of mobile public service capacity. This issue is especially important at the provincial level, where governments differ in fiscal resources, digital infrastructure, administrative priorities, leadership attention, public expectations, and exposure to peer competition. Because integrated digital public service delivery increasingly depends on mobile platforms, these differences create a need to move beyond isolated determinants and examine the combinations of conditions through which local governments build service capacity. To address these theoretical and empirical gaps, this study examines provincial mobile government practices through an analytical framework that integrates institutional pressure with organizational support and evaluates their joint explanatory relevance from a configurational perspective.

2.2 Theoretical Foundations: Institutional Pressure and Organizational Support

Neo-institutional theory holds that organizations are embedded in institutional environments that generate pressures for conformity and influence the rationalization of organizational structures and behavior, a process generally conceptualized as institutional pressure [24]. DiMaggio and Powell [25] distinguished three principal mechanisms through which such pressure operates: coercive, mimetic, and normative pressure. Early formulations of neo-institutional theory primarily emphasized organizational isomorphism, suggesting that organizations exposed to similar institutional environments tend to become increasingly alike. However, persistent behavioral heterogeneity among organizations indicates that common institutional pressures do not necessarily produce uniform responses. Later scholarship therefore recognized organizations as open systems embedded in multiple institutional logics, whose interaction can generate differentiated interpretations and strategic responses [26]. This shift from explaining convergence toward explaining variation is particularly relevant to mobile government services, because provincial governments operate within a shared national institutional system yet display substantial differences in digital public service capacity. During policy implementation, local governments are exposed simultaneously to vertical policy requirements, horizontal comparison with peer jurisdictions, and expectations arising from citizens and the broader social environment. Vertical policy transmission creates accountability and compliance pressures, horizontal comparison creates competitive and learning pressures, while public and media attention can generate continuing expectations for improved service performance. These pressures also interact with differences in resource availability and policy priorities. Because institutional pressures operate within fields characterized by multiple and sometimes competing logics, they can generate differentiated implementation strategies rather than uniform organizational behavior [27]. In this study, digital public service implementation is therefore conceptualized as an organizational response to external institutional conditions, while the interpretation of policy demands and the mobilization of local resources constitute the internal basis through which those pressures are converted into action. As illustrated in Figure 1, this relationship involves three connected processes: institutional pressures transmitted through the governance system can generate heterogeneous organizational responses; differences in policy interpretation and resource endowments influence how governments allocate technical, financial, and leadership support; and the

interaction of these external and internal conditions produces differentiated levels of mobile government service capacity across jurisdictions.

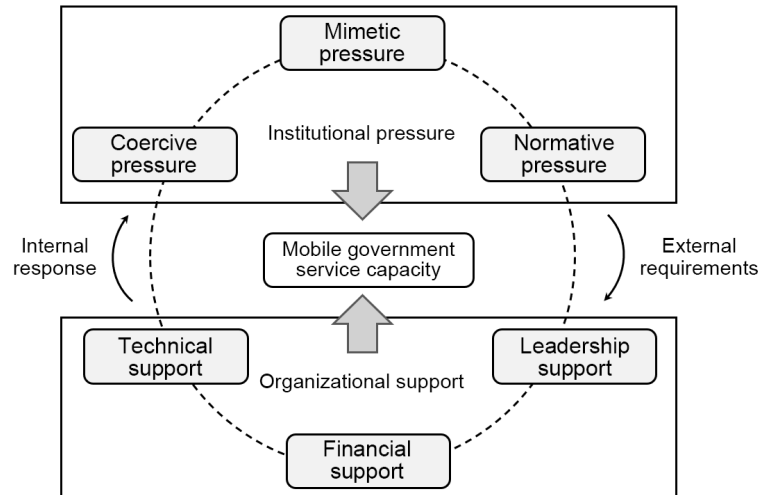


Figure 1. Theoretical foundations of this study

Although institutional pressure frameworks were initially developed largely in studies of private organizations, corporate behavior, and innovation, subsequent research has demonstrated their relevance to the diffusion of policy and public service innovation among local governments [28]. As institutional actors, local governments do not develop mobile government services independently of the broader governance environment. Their decisions are shaped simultaneously by top-down policy direction, horizontal comparison and competition, and bottom-up social expectations. Coercive pressure arises from formal rules, laws, administrative mandates, regulatory standards, and other requirements that create obligations for organizational compliance [24]. In the context of digital public services, provincial governments translate national directives into local policies, implementation rules, and technical arrangements, while both central mandates and provincial regulations influence the allocation of resources to mobile service development. Mimetic pressure emerges under conditions of uncertainty or competitive comparison, encouraging governments to observe and emulate jurisdictions perceived as successful. For provincial governments, such pressure can arise from performance comparisons, regional competition, and the reputational consequences of lagging behind neighboring or benchmark jurisdictions. Normative pressure originates from professional expectations, social standards, and broader beliefs concerning appropriate organizational conduct [25]. In digital governance, rising public expectations for accessible and efficient services can gradually establish standards against which government performance is judged; as neighboring regions develop more advanced mobile services, citizens' expectations may also increase and become an additional force encouraging service modernization [18].

Institutional pressure alone, however, does not determine implementation outcomes. As external expectations move through administrative systems, governments retain substantial discretion in how they interpret demands, prioritize policy objectives, and mobilize organizational resources, producing heterogeneous responses even under broadly comparable institutional conditions [26]. Governments may place different issues at different positions within their policy agendas and consequently devote unequal levels of attention and support to their implementation [29]. On this basis, the present study conceptualizes organizational support through three dimensions: technical, financial, and leadership support. Technical support encompasses the digital infrastructure, technological expenditure, and related capabilities required to operate mobile public service systems; these resources influence service efficiency, system quality, and the capacity to support user adoption [2, 17]. Financial support provides the fiscal resources necessary for platform development, operation, maintenance, system upgrading, and external service procurement, making sustained expenditure an important condition for maintaining digital service capacity [30]. Leadership support reflects the degree to which senior administrative actors assign attention, authority, and organizational priority to digital government initiatives. Strong leadership can elevate complex digital projects within the formal policy agenda, facilitate interdepartmental coordination, mobilize organizational resources, and improve implementation efficiency [16].

Taken together, institutional pressure and organizational support represent two analytically distinct but interdependent dimensions of mobile government service capacity. Institutional pressures define the external expectations, incentives, and constraints under which provincial governments operate, whereas organizational support reflects the internal capacity and strategic commitment through which governments respond to those conditions. Their relationship

is unlikely to be purely additive. Similar institutional pressures may generate different service outcomes when organizational resources differ, while strong organizational support may compensate for the absence or weakness of particular external pressures. Conversely, favorable institutional conditions may fail to produce high service capacity when essential organizational resources are absent. A configurational perspective is therefore appropriate for examining how different combinations of coercive, mimetic, and normative pressure interact with technical, financial, and leadership support, as well as whether alternative combinations can lead to comparable levels of mobile government service capacity. This theoretical logic provides the basis for the empirical analysis that follows.

3 Data Source and Research Method

3.1 Sample Selection

The analysis covers the mobile government service platforms of all 31 provincial-level governments in China, with the outcome data drawn from the China Provincial Mobile Government Service Report 2023 [31]. Provincial governments provide an appropriate level of analysis because they occupy an intermediate position in China's multilevel governance system. They are responsible for translating national digital government policies into provincial implementation arrangements while simultaneously coordinating the development and delivery of mobile government services across lower-level jurisdictions. This dual role makes provincial governments important organizational actors linking national institutional requirements with locally differentiated administrative resources and implementation strategies. All 31 provincial-level governments included in the study had established their own mobile government service applications, allowing the analysis to compare jurisdictions operating within a common national policy framework but under heterogeneous institutional, fiscal, technical, and organizational conditions. The complete provincial sample therefore provides an appropriate empirical basis for examining configurational differences in mobile government service capacity across China.

3.2 Research Method

A configurational perspective was adopted to examine how institutional pressures and organizational support jointly shape mobile government service capacity at the provincial level. Fuzzy-set qualitative comparative analysis (fsQCA) was employed because the research focuses on combinations of conditions rather than the isolated net effects of individual variables. The method is also appropriate for the 31 provincial-level cases included in the analysis, as it is well suited to medium-sized samples and to research designs concerned with causal complexity, conjunctural causation, and multiple pathways to the same outcome.

Three considerations informed the methodological choice. First, the theoretical framework assumes that coercive, mimetic, and normative pressures interact with technical, financial, and leadership support rather than exerting independent effects on mobile government service capacity. QCA enables these conditions to be examined as configurations and is therefore better aligned with the study's theoretical interest in their joint operation than approaches centered exclusively on net effects; it also reduces problems associated with examining highly interdependent explanatory conditions through conventional additive models [32]. Second, Chinese provincial governments differ substantially in their institutional environments, resource endowments, and implementation strategies. QCA allows alternative combinations of conditions to be identified that may lead to comparable levels of mobile government service capacity, thereby capturing equifinality and revealing differentiated governance pathways. Third, the antecedent conditions vary by degree rather than conforming to simple binary distinctions. The fuzzy-set form of QCA is therefore appropriate because it allows cases to hold partial membership in theoretically defined sets while preserving the configurational structure of the analysis.

3.3 Variable Specification and Calibration

3.3.1 Outcome variable

Mobile government service capacity is the outcome variable and is operationalized using the China Provincial Mobile Government Service Report 2023 [31]. It refers to the capacity of provincial governments to provide accessible, functional, user-friendly, and trustworthy public services through mobile government platforms. The report evaluates provincial mobile government applications and related government service mini-programs across five dimensions: availability, utility, usability, appeal, and trustworthiness. These dimensions collectively capture the extent to which mobile platforms provide usable and effective channels for digital public service delivery. The resulting "Easy-to-Handle" index assigns grades ranging from A+ to B—, which are converted into scores from 6 to 1, respectively. This index is used as the outcome measure for subsequent fuzzy-set calibration and configurational analysis.

3.3.2 Antecedent variables

Coercive pressure. Coercive pressure is operationalized by the number of provincial policy documents related to digital government and government services. The Peking University Law Database [33] was searched for documents

classified as “local regulations” or “local normative documents” whose titles or contents contained terms including “mobile government,” “e-government,” and “government services.” The retrieved documents were screened for contextual relevance and current validity. The resulting number of valid policy documents for each province is used to represent the intensity of formal regulatory and policy pressure associated with mobile government development.

Mimetic pressure. Mimetic pressure is operationalized through the mobile government service performance of geographically neighboring provinces. For each province, the average mobile government service capacity score of its neighboring provinces is calculated. Higher neighboring-province performance indicates greater exposure to high-performing peer governments and, consequently, stronger incentives for intergovernmental learning, comparison, and imitation. This measure captures the competitive and demonstrative pressures that neighboring jurisdictions may exert on the focal province’s mobile government service development.

Normative pressure. Normative pressure reflects societal expectations and public demand for digital government services. As mobile internet use expands, citizens are increasingly able to access digital services and develop expectations regarding the accessibility, responsiveness, and convenience of government service provision. The proportion of mobile internet users in each province is therefore used as a proxy for the intensity of normative pressure associated with digital service demand. Provincial mobile internet user data were extracted from the China Statistical Yearbook 2023 [34].

Technical support. Technical support is measured as the proportion of information technology expenditure in general fiscal expenditure. This indicator reflects the relative fiscal commitment of provincial governments to the technological resources required for digital public service provision. Because the effects of technology investment may not be realized immediately, the average proportion of information technology expenditure over 2021–2023 is used to reduce the influence of short-term annual fluctuations and better represent sustained technical support. The data were obtained from the China Statistical Yearbook 2023 [34].

Financial support. Financial support is operationalized as the proportion of general public service expenditure in each province’s local fiscal expenditure. The indicator captures the relative allocation of fiscal resources to public administration and service provision and is used to represent the financial support available for mobile government development. A higher proportion indicates a stronger fiscal commitment to the administrative and service functions associated with digital public service delivery. The corresponding provincial fiscal expenditure data were sourced from the China Statistical Yearbook 2023 [34].

Leadership support. Leadership support is assessed using three observable indicators: whether a dedicated leading working group has been established for “Internet Plus Government Services” or digital government development; whether the group is headed by the provincial governor or a deputy governor; and whether relevant conferences, symposia, or training activities have been organized. These indicators capture formal organizational arrangements, senior-level involvement, and sustained managerial attention to digital government initiatives. Equal weights were assigned because the three indicators represent distinct dimensions of leadership engagement and no empirical basis was available for assigning differential weights. A composite score ranging from 0 to 3 was then constructed according to the number of criteria satisfied: 3 points for all three criteria, 2 points for two criteria, 1 point for one criterion, and 0 points when none of the criteria was satisfied.

Table 1. Descriptive statistics of variables and calibration points

Variable	Descriptive Statistics				Calibration Threshold		
	Mean	SD	Min	Max	Full Membership	Crossover Point	Full Non-Membership
Policy quantity	51.613	29.430	5.000	134.000	107.500	58.400	11.500
Government capacity index at the same level	2.985	0.772	1.750	4.667	4.367	3.227	1.875
Proportion of mobile Internet users	8.179	0.838	5.707	9.662	9.168	8.433	6.573
Information technology expenditure	0.024	0.016	0.004	0.062	0.054	0.028	0.006
Public service expenditure	0.084	0.013	0.048	0.120	0.101	0.087	0.065
Importance of leaders	2.129	0.793	1	3	3.000	2.900	1.000
Mobile government service evaluation index	3.000	1.295	1	6	5.500	3.000	1.500

Note: SD indicates standard deviation. Calibration thresholds were determined using the direct calibration method, with full membership, crossover, and full non-membership thresholds set at the 95th percentile, median, and 5th percentile values, respectively.

3.3.3 Variable calibration

All variables were transformed into fuzzy-set membership scores using the direct calibration method. For each variable, the thresholds for full membership, the crossover point, and full non-membership were set at the 95th percentile, median, and 5th percentile of its empirical distribution, respectively. This procedure provides consistent calibration criteria across the conditions while retaining the relative differences among the 31 provincial cases. To prevent cases located exactly at the maximum-ambiguity membership score of 0.500 from being excluded from subsequent set-theoretic comparisons, such scores were adjusted to 0.499. Calibration was performed using fsQCA 4.1. Table 1 reports the descriptive statistics and calibration anchors for the outcome and antecedent conditions.

4 Result Analysis

4.1 Analysis of Single Antecedent Conditions

A necessity analysis was first conducted to determine whether any individual antecedent condition, or its absence, constituted a necessary condition for high or non-high mobile government service capacity. Table 2 reports the consistency and coverage values for each condition. Among the conditions associated with high mobile government service capacity, high mimetic pressure and strong technical support showed comparatively high consistency and coverage, indicating relatively strong set-theoretic relationships with the outcome. However, the consistency of every individual condition remained below the conventional necessity threshold of 0.90. No single institutional pressure or dimension of organizational support can therefore be regarded as necessary for high mobile government service capacity. This result supports the use of configurational analysis to examine how multiple conditions operate jointly in producing different levels of digital public service capacity.

Table 2. Necessity analysis of antecedent conditions

Condition Variable	High Mobile Government Service Capacity		Non-High Mobile Government Service Capacity	
	Consistency	Coverage	Consistency	Coverage
High coercive pressure	0.595	0.634	0.531	0.746
Low coercive pressure	0.762	0.552	0.739	0.706
High mimetic pressure	0.742	0.788	0.402	0.563
Low mimetic pressure	0.588	0.427	0.848	0.812
High normative pressure	0.675	0.590	0.572	0.659
Low normative pressure	0.609	0.519	0.644	0.723
Strong technical support	0.708	0.794	0.334	0.494
Weak technical support	0.549	0.385	0.861	0.795
Strong financial support	0.595	0.572	0.569	0.722
Weak financial support	0.711	0.556	0.663	0.683
Strong leadership support	0.572	0.551	0.484	0.614
Weak leadership support	0.599	0.468	0.646	0.666

Note: Consistency and coverage values are calculated based on the necessity analysis of antecedent conditions.

4.2 Configurational Analysis

Before Boolean minimization, the case-frequency threshold was set to 1, the sufficiency consistency threshold to 0.80, and the proportional reduction in inconsistency threshold to 0.70 [35]. Because the necessity analysis identified no individual condition as necessary for mobile government service capacity, no directional simplifying assumptions were imposed in the counterfactual analysis; both the presence and absence of each antecedent condition were therefore allowed to contribute to configurations associated with high or non-high capacity. The prime implicant algorithm was subsequently used to minimize the truth table and derive the configurational solutions.

As reported in Table 3, five configurations were associated with high mobile government service capacity. On the basis of their core and peripheral conditions, these configurations were grouped into three broader pathways: a coercive-pressure-led pathway, a mimetic/normative-pressure-driven pathway, and a three-pressure-linkage pathway. The overall solution consistency for the five high-capacity configurations was 0.885, indicating a strong set-theoretic relationship between the identified configurations and high mobile government service capacity. Their overall coverage was 0.631, meaning that the configurations jointly covered 63.1% of membership in the high-capacity outcome set. Configurations associated with non-high mobile government service capacity were also examined to capture the causal asymmetry inherent in the configurational approach.

Table 3. Results of configuration analysis

Conditional Variables	High Mobile Government Service Capacity					Non-High Mobile Government Service Capacity		
	H1a	H1b	H2a	H2b	H3	H4a	H4b	H5
Coercive pressure	●	●	⊗	⊗	●	●	●	⊗
Mimetic pressure	●	⊗	●	●	●	⊗	⊗	⊗
Normative pressure		●	⊗	●	●		●	⊗
Technical support	●	●	●	●	●	⊗	⊗	
Financial support	⊗	●	⊗	●			⊗	●
Leadership support	●	⊗	●	⊗	●	⊗		●
Raw coverage	0.308	0.176	0.301	0.208	0.363	0.321	0.281	0.239
Unique coverage	0.018	0.024	0.144	0.052	0.067	0.108	0.024	0.074
Consistency	0.941	0.943	0.889	0.853	0.875	0.988	0.959	0.934
Representative case	Anhui Jiangxi	Guangdong	Shanghai Fujian	Zhejiang	Jiangsu	Gansu Yunnan Guizhou Hebei	Guangxi Sichuan	Xizang Shanxi
Overall coverage	0.631					0.467		
Overall consistency	0.885					0.961		

Note: ● represents core conditions, ● represents marginal conditions, ⊗ indicates the absence of a condition, and blank cells indicate irrelevant conditions.

4.2.1 Coercive-pressure-led pathway

The first pathway is coercive-pressure-led and comprises configurations H1a and H1b. Coercive pressure is present as a core condition in both configurations, indicating that formal policy and regulatory pressure provides the principal institutional basis for these two routes to high mobile government service capacity. However, coercive pressure does not operate independently. Its association with high capacity depends on different combinations of other institutional pressures and organizational support conditions, illustrating how similar policy mandates can be translated into high service capacity through different provincial governance arrangements.

In H1a, strong coercive pressure is combined with mimetic pressure and leadership support. Clear policy requirements from central and provincial governments establish formal implementation expectations, while the performance of neighboring provinces creates additional competitive and learning pressure. Leadership attention provides the organizational support needed to translate these pressures into implementation. H1a has a raw coverage of 0.308 and is represented by Anhui and Jiangxi. Jiangxi illustrates this configuration. In 2017, Jiangxi advanced its digital government development under the broader institutional framework established by the “Internet Plus Government Services” initiative [36]. Located near comparatively strong performers such as Zhejiang and Anhui, Jiangxi drew on peer experience and introduced an implementation plan centered on the “One Network, Three Platforms” initiative [37]. Provincial leaders also received dedicated briefings, the executive vice-governor headed the “Three Lists, One Network” working group, and a Government Services Management Office was established to coordinate implementation. Jiangxi subsequently maintained a Grade A mobile government service assessment [31].

H1b represents an alternative route under the same core condition of coercive pressure. Where mimetic pressure from neighboring provinces is weaker, normative pressure and financial support combine with formal policy requirements to support high mobile government service capacity. H1b has a raw coverage of 0.176 and is represented by Guangdong. Guangdong combines substantial fiscal capacity with extensive formal policy engagement in digital government. Since the development of mobile government services began, the province has issued 103 local normative documents related to “Internet Plus Government Services,” the highest number among the provincial cases examined [33]. The Guangdong Provincial People’s Congress Standing Committee also adopted the Guangdong Province Government Service Digitalization Regulations establishing a provincial regulatory framework for digital public service delivery [38]. In this configuration, strong coercive pressure, public demand, and financial support jointly characterize Guangdong’s pathway to high mobile government service capacity.

4.2.2 Mimetic/normative-pressure-driven pathway

The second pathway comprises configurations H2a and H2b and is characterized by the absence of strong coercive pressure. Under these configurations, high mobile government service capacity is associated instead with mimetic pressure, either in combination with leadership support or together with normative pressure and financial support.

The two configurations therefore show how provincial governments can achieve strong digital public service capacity even when formal policy pressure is not the dominant institutional force.

H2a combines strong mimetic pressure with leadership support and technical support. Under this configuration, competitive and learning pressures generated by neighboring governments are accompanied by sustained leadership attention, providing an alternative route to high mobile government service capacity despite the absence of strong coercive and normative pressure. H2a has a raw coverage of 0.301 and is represented by Shanghai and Fujian. Fujian provides an illustrative case. The province developed its e-government infrastructure relatively early and had established an online government affairs hall by late 2014 [39]. This early development occurred before the broader national promotion of “Internet Plus Government Services,” which later strengthened the institutional environment for digital government development [36]. A “Digital Fujian” leading working group headed by the Provincial Party Committee Secretary was subsequently established, and the province continued to institutionalize digital development through initiatives including the 2024 Digital Fujian Work Plan [40]. These measures illustrate how peer-oriented development and sustained leadership attention can accompany strong mobile government service performance.

H2b differs from H2a by combining mimetic pressure with normative pressure and financial support. High-performing neighboring jurisdictions and strong public demand for digital services create external incentives, while fiscal support provides the organizational resources required to respond to those pressures. H2b has a raw coverage of 0.208 and is represented by Zhejiang. Zhejiang operates within a highly developed regional economy and borders several provinces with relatively strong digital government performance. Its substantial economic and fiscal resources have supported continued investment in digital public services [34]. The province has long served as a benchmark for “Internet Plus Government Services,” and its mobile government service assessment reached the A+ level [31]. In 2021, the provincial government procured an “Internet Plus Government Services Platform Construction” project for the Provincial Big Data Development and Management Bureau. More recently, Hangzhou has developed value-added government service platforms, further extending the province’s digital service arrangements.

4.2.3 Three-pressure-linkage pathway

The third pathway corresponds to configuration H3 and is distinguished by the simultaneous presence of all three forms of institutional pressure. Coercive pressure appears as a core condition, while mimetic pressure, normative pressure, technical support, and leadership support are jointly present in the configuration. H3 has the highest raw coverage among the five high-capacity configurations, at 0.363, indicating that it covers a comparatively large share of membership in the high mobile government service capacity set.

H3 demonstrates a pathway in which formal policy requirements, intergovernmental comparison, public expectations, and organizational support operate concurrently. Within China’s multilevel governance system, “Internet Plus Government Services” requirements are transmitted through the administrative hierarchy, while performance comparison and peer competition provide additional incentives for provincial governments to improve digital services. At the same time, growing public expectations generate normative pressure, and technical and leadership support provide organizational capacity for implementation. High mobile government service capacity under this configuration is therefore associated with the alignment of multiple external institutional pressures and internal organizational support conditions.

Jiangsu is the representative case. In 2015, the Central Work Report called for the advancement of e-government [41], and the Jiangsu Provincial Government established the Jiangsu Province Government Services Center in Nanjing in the same year. In 2016, the province established a Jiangsu Government Service Network Construction Leading Working Group and issued implementation guidelines for network development [42]. As one of China’s most populous provinces, Jiangsu also faces substantial demand for accessible government services. The Jiangsu Government Service App and Jiangsu E-Government External Network were launched in 2017, followed by government service channels through Alipay and WeChat mini-programs. In 2023, the Provincial Government Affairs Office announced plans to complete a province-wide “One Cloud” system by the end of 2027, and in 2024 Jiangsu launched a province-wide integrated hotline service platform. The case illustrates how multiple institutional pressures can coincide with technical and leadership support in a high-capacity provincial digital service system.

4.2.4 Substitution and complementarity across configurational pathways

Cross-configuration comparisons provide further insight into how institutional pressures and organizational support operate across alternative pathways. Two patterns are particularly evident. First, coercive and mimetic pressures recur across several high-capacity configurations, although neither is present in every pathway. Their repeated appearance suggests that external institutional expectations constitute important components of several routes to high mobile government service capacity, while the absence of a universally shared institutional-pressure condition reinforces the equifinal character of the results. Second, strong technical support is present across all five high-capacity configurations. This recurring pattern indicates that technical capacity is a particularly prominent component of the identified high-capacity pathways, although the necessity analysis shows that it should not be interpreted as an individually necessary condition.

Comparisons within the three pathway types also reveal potential substitution among conditions. Within the mimetic/normative-pressure-driven pathway, comparison of H2a and H2b indicates that, under strong mimetic pressure, leadership support in H2a and the combination of normative pressure and financial support in H2b represent alternative organizational-institutional arrangements associated with high capacity. A similar pattern appears within the coercive-pressure-led pathway. Comparison of H1a and H1b suggests that mimetic pressure combined with leadership support and normative pressure combined with financial support can occupy alternative positions under strong coercive pressure. Where intergovernmental competition and learning are pronounced, leadership attention can accompany the translation of external pressure into implementation. Where such horizontal pressure is weaker, stronger public demand combined with financial support provides an alternative configuration associated with high service capacity. These comparisons indicate that provincial governments need not rely on an identical set of institutional and organizational conditions; different combinations can support comparable digital public service outcomes.

4.3 Asymmetric Pathways to Non-High Mobile Government Service Capacity

The analysis of the negated outcome further demonstrates the causal asymmetry of mobile government service capacity. As shown in Table 3, three configurations—H4a, H4b, and H5—are associated with non-high capacity. Across the three configurations, weak technical support and low mimetic pressure recur as core conditions, although their specific roles vary across configurational pathways. The configurations leading to non-high capacity are not simple inversions of those associated with high capacity, confirming that the mechanisms underlying stronger and weaker digital public service performance are asymmetric.

Comparison of H4a and H4b further shows that the presence of normative pressure or financial support does not necessarily correspond to high capacity when important organizational conditions are absent. In particular, weak leadership support appears as a core condition in H4a despite the presence of other potentially favorable conditions. Comparison with H5 similarly indicates that limited external institutional pressure combined with insufficient organizational support can characterize a distinct route to non-high mobile government service capacity. Taken together, these configurations reinforce the central configurational finding of the study: service capacity is associated with combinations of institutional and organizational conditions rather than the presence or absence of any single factor.

4.4 Robustness Check

The robustness of the configurational results was assessed by varying key analytical thresholds [43]. First, the case-frequency threshold was increased from 1 to 2. The resulting intermediate solution was a subset of the original solution and retained the principal configurational structure. Second, the sufficiency consistency threshold was increased from 0.80 to 0.90. The resulting configurations were likewise subsets of the original solutions. These tests indicate that the principal configurational patterns remain stable under more restrictive analytical thresholds, supporting the robustness of the empirical results.

5 Discussion

This study adopts a configurational perspective to explain how institutional pressures and organizational support combine to shape mobile government service capacity across Chinese provinces. The identification of three broad pathways—the coercive-pressure-led, mimetic/normative-pressure-driven, and three-pressure-linkage pathways—shows that high digital public service capacity can emerge from multiple institutional and organizational arrangements rather than from a single dominant condition. The coercive-pressure-led pathway demonstrates the continuing importance of formal policy requirements within China's multilevel governance system, while the other pathways show that comparable outcomes can also arise through different combinations of peer pressure, public expectations, and organizational support. This finding is consistent with recent evidence that central policy mandates and performance-oriented implementation arrangements remain important influences on local digital government development in China [44]. At the same time, the recurring presence of mimetic pressure across several high-capacity configurations extends this interpretation beyond hierarchical policy transmission. Rather than functioning solely as a response to uncertainty, mimetic pressure in the provincial digital governance context can also reflect intergovernmental comparison, policy learning, and competition among jurisdictions. This interpretation is consistent with research showing that performance comparison and competitive incentives can shape the behavior of Chinese local governments [45]. The results therefore suggest that provincial digital public service development is embedded in a multilevel institutional environment in which vertical policy requirements and horizontal intergovernmental relationships can operate through different configurational pathways.

The findings also clarify the conditional roles of normative pressure and technical support. Normative pressure does not constitute a necessary condition for high mobile government service capacity, nor does it appear independently as a sufficient explanation of the outcome. Instead, it occurs within configurations that also contain other institutional

pressures or organizational resources. This pattern indicates that public expectations for accessible and effective digital services are more likely to be associated with high service capacity when governments possess complementary conditions that allow such expectations to be translated into implementation. In some configurations, financial support accompanies normative pressure, providing resources for platform development and service provision; in others, normative pressure operates alongside a broader combination of coercive and mimetic pressures. This interpretation is consistent with research emphasizing the interaction between institutional demand and organizational capacity in digital government development [46]. Technical support displays a different configurational pattern. Although the necessity analysis does not identify it as an individually necessary condition, strong technical support is present in all five configurations associated with high mobile government service capacity. Its recurrence across otherwise different pathways indicates that technological resources form a particularly important component of the organizational capacity through which institutional demands are translated into digital public service provision. This finding is consistent with evidence that technological resources and organizational capabilities provide important foundations for improving digital government performance under institutional pressures [47].

A further theoretical contribution concerns the potential substitution and complementarity among institutional and organizational conditions. The comparison of configurations indicates that high mobile government service capacity does not require an identical combination of leadership, financial resources, and external pressures across all provinces. Under strong mimetic or coercive pressure, leadership support in some configurations occupies a position comparable to the combination of normative pressure and financial support in others. This pattern illustrates the equifinality of digital governance pathways and differs from additive explanations in which favorable organizational conditions are expected to accumulate in a uniform manner. Instead, the results point to context-dependent combinations of institutional demands and organizational resources, consistent with broader research emphasizing the importance of institutional and organizational context in local government technology adoption [48]. Provinces with relatively strong fiscal resources and public demand may rely more heavily on resource-supported implementation, whereas provinces exposed to stronger intergovernmental comparison may depend more on leadership attention and organizational coordination. These should be understood as alternative configurational patterns rather than mutually exclusive governance models. Leadership support can facilitate agenda setting, interdepartmental coordination, and targeted resource mobilization, while financial support can provide the material basis for sustained platform development. Such variation is consistent with evidence that local digital transformation depends on how administrative leadership and organizational capacity are aligned with particular institutional environments [49].

Taken together, these findings extend the understanding of digital public service capacity in two respects. Theoretically, they shift attention from the isolated effects of individual institutional or organizational determinants toward the combinations through which external pressures and internal support jointly correspond to service outcomes. Empirically, they demonstrate that provincial governments operating within the same national governance system can reach high mobile government service capacity through different combinations of institutional and organizational conditions. This configurational heterogeneity is particularly relevant to place-based governance because differences in fiscal capacity, technical resources, public demand, leadership attention, and exposure to neighboring jurisdictions shape the institutional contexts in which digital public services are implemented. The findings therefore provide a more differentiated explanation of digital service capacity across local governance contexts and reinforce the importance of aligning institutional demands with locally available organizational resources.

6 Conclusions

This study examined how external institutional pressures and internal organizational support combine to shape mobile government service capacity across Chinese provinces. By integrating coercive, mimetic, and normative pressures with technical, financial, and leadership support within a configurational framework, the analysis identified multiple pathways associated with high and non-high levels of digital public service capacity.

Three broad pathways were associated with high mobile government service capacity: a coercive-pressure-led pathway, a mimetic/normative-pressure-driven pathway, and a three-pressure-linkage pathway. The results demonstrate clear equifinality, indicating that provincial governments can achieve comparable high-capacity outcomes through different combinations of institutional and organizational conditions. No individual antecedent condition met the threshold for a necessary condition. Nevertheless, strong technical support was present across all five high-capacity configurations, highlighting its recurring role within otherwise heterogeneous pathways. Cross-configuration comparisons further revealed potential substitution between leadership support and combinations of normative pressure and financial support under different institutional conditions.

The analysis of non-high mobile government service capacity further demonstrated causal asymmetry. Weak technical support and low mimetic pressure appeared as recurring core conditions across the identified non-high-capacity configurations, while weak leadership support characterized one pathway even when other favorable conditions were present. The pathways associated with non-high capacity were therefore not simple inversions of those associated with high capacity.

Overall, the findings demonstrate that differences in mobile government service capacity cannot be adequately explained by institutional pressure, technological resources, financial investment, or leadership attention in isolation. Instead, digital public service capacity is associated with the alignment of external institutional environments and internal organizational support under heterogeneous provincial governance conditions. This configurational perspective provides a basis for understanding why governments operating within a common national policy framework can follow different routes to effective digital public service delivery.

7 Policy Implications

The configurational findings have several implications for provincial and local authorities seeking to strengthen mobile government service capacity. First, institutional pressures should be treated as differentiated governance instruments rather than as interchangeable sources of administrative pressure [14]. The results indicate that coercive pressure can form an important component of high-capacity pathways, but formal policy requirements alone do not guarantee strong digital service performance. Clear policy frameworks and implementation responsibilities should therefore be accompanied by mechanisms that improve the transparency and evaluation of public service performance. Consistent with research emphasizing the role of transparency in shaping citizens' responses to e-government services [18], publicly accessible performance information and systematic service evaluation can strengthen the connection between institutional expectations and service improvement. The practical objective is not simply to intensify administrative pressure, but to ensure that different forms of institutional expectation are translated into clear implementation priorities and measurable service outcomes.

Second, local governments should strengthen the alignment between external institutional demands and internal organizational capacity [10]. The results show that institutional pressure is associated with high service capacity only within particular combinations of organizational support. Leadership attention is especially relevant where digital initiatives require coordination across departments, administrative levels, and service systems. Consistent with evidence on collaborative leadership in e-government transformation [16], provincial and local authorities can use formal coordinating bodies, clearly assigned responsibilities, and sustained senior-level oversight to reduce organizational fragmentation and facilitate implementation. At the same time, the recurrence of technical support across all identified high-capacity configurations highlights the importance of maintaining adequate technological resources for platform operation, integration, and service delivery. Institutional objectives should therefore be matched with the technical and managerial capacity required to implement them.

Third, policy strategies should recognize that alternative combinations of conditions may support comparable service outcomes rather than assuming that a single governance model is appropriate for every jurisdiction [27]. Provinces differ substantially in fiscal resources, technical capacity, public demand, leadership arrangements, and exposure to intergovernmental competition. Resource-constrained governments should therefore identify which organizational conditions can realistically be strengthened within their local context rather than mechanically reproducing the institutional arrangements of better-resourced jurisdictions. Where horizontal intergovernmental comparison is relatively weak, stronger financial and organizational support may help compensate for the absence of peer-driven incentives. Conversely, where fiscal resources are constrained, leadership attention, administrative coordination, and policy learning from comparable jurisdictions may provide alternative forms of organizational support. Sustained financial investment nevertheless remains important for the development and maintenance of digital service infrastructure, consistent with evidence on the resource requirements of public-sector innovation [30]. A differentiated approach to resource allocation and institutional coordination is therefore more consistent with the configurational findings than a uniform model of digital government development.

8 Limitations and Future Research

Several limitations should be considered when interpreting the findings and provide directions for future research. First, the analytical framework focuses on institutional pressure and organizational support and therefore does not capture all factors that may shape mobile government service capacity. Future research could incorporate complementary perspectives, including behavioral organization theory and resource dependence theory, to examine additional mechanisms through which governments interpret institutional demands, acquire resources, and make implementation decisions.

Second, the study relies primarily on cross-sectional provincial-level data. The configurational relationships identified here should therefore not be interpreted as evidence of temporal causal sequences. Institutional pressures, organizational support, and mobile government service capacity may evolve together as digital governance systems mature. Longitudinal or panel-based configurational designs could examine how provincial pathways change over time and whether governments move between different combinations of institutional and organizational conditions during successive stages of digital transformation.

Third, several antecedent conditions are represented by proxy indicators derived from available statistical and administrative data. Although these indicators provide a consistent basis for comparison across the 31 provincial-level

cases, they cannot fully capture the multidimensional nature of institutional pressure, technical capacity, financial commitment, or leadership support. Future studies could combine administrative indicators with survey data, policy-text analysis, interviews, or platform-level performance measures to improve construct validity and provide a more detailed account of the mechanisms underlying different configurations. Comparative research across countries or across municipal and metropolitan levels could also help determine whether the configurational patterns identified in the Chinese provincial context extend to other institutional and local governance settings.

Author Contributions

Conceptualization, G.D.W. and Q.W.L.; methodology, Q.W.L.; software, Q.W.L. and Z.B.H.; validation, X.B.Z., J.L.H., and G.D.W.; formal analysis, Q.W.L.; investigation, G.D.W. and Q.W.L.; resources, G.D.W. and Q.W.L.; data curation, G.D.W. and Q.W.L.; writing—original draft preparation, G.D.W., Q.W.L., X.B.Z., and J.L.H.; writing—review and editing, G.D.W., Q.W.L., Z.B.H., X.B.Z., and J.L.H.; visualization, Q.W.L. and Z.B.H.; supervision, G.D.W.; project administration, G.D.W.; funding acquisition, G.D.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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